

Coding Assignments #6 & #7 (Part One)

August 4, 2025

Directions:

1. You will be penalized for not following each of the directions exactly as written below. If you aren't sure about what something means, ask the professor!
 2. Put all your work in a single Haskell file.
 3. Submit your work (before the deadline) to Canvas.
 4. **Important:** You must avoid global helper functions. Keep the scope of your helper functions limited by using `lets` and `wheres`.
 5. **Important:** You will be graded on excessive use of helper functions when patterns or guards would make more sense.
 6. **Important:** You will lose points **each time** you use the `if` keyword. Use patterns and guards instead!
1. (Call your datatype **Mat2x2** and constructor: **Matrix**). Using only techniques from chapters 1-7a, make a datatype for a 2×2 matrix. You should use a type parameter so that matrices can be polymorphic. Your constructor should simply list 4 values: the top-left, top-right, bottom-left, and bottom-right values of the matrix.
 2. (Call your function **transpose**). Using only techniques from chapters 1-7a, write a function that computes the transpose of the matrix. The only thing that changes in the transpose of a matrix is that the bottom-left and top-right values switch positions.
 3. (Call your function **determinant**). Using only techniques from chapters 1-7a, write a function that computes the determinant of a 2×2 matrix. The formula for determinant is:

$$(\text{top-left} \times \text{bottom-right}) - (\text{bottom-left} \times \text{top-right}).$$

4. (Call your function **inverse**). Using only techniques from chapters 1-7a, write a function that computes the inverse of a 2x2 matrix. The formulae for computing the inverse are:

$$(\text{new}) \text{ top-left} = \frac{1}{\text{determinant}} \times \text{bottom-right} \quad (1)$$

$$(\text{new}) \text{ top-right} = \frac{1}{\text{determinant}} \times -(\text{top-right}) \quad (2)$$

$$(\text{new}) \text{ bottom-left} = \frac{1}{\text{determinant}} \times -(\text{bottom-left}) \quad (3)$$

$$(\text{new}) \text{ bottom-right} = \frac{1}{\text{determinant}} \times \text{top-left} \quad (4)$$

The values on the right-hand side of the equals sign in each formula are the original matrix values, the values on the left hand side are the new matrix values.

5. (Call your function **matMult**). Using only techniques from chapters 1-7a, write a function that takes two arguments: a **Mat2x2** matrix and a vector, i.e., a pair of values (**u**, **v**). The function computes the matrix multiplied by the vector. The formulas for the output vector are

$$(\text{top-left} \times u + \text{top-right} \times v, \text{bottom-left} \times u + \text{bottom-right} \times v)$$

6. (Call your function **solve**). Using only techniques from chapters 1-7a, write a function that solves the matrix equation $Ax = b$, where A is a **Mat2x2** and B is a vector, i.e., a pair of values (**u**, **v**). The formula for the solution is $x = A^{-1} \times b$, where " A^{-1} " means the inverse of A and " $\times$ " means matrix-multiply.